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Sustainable AI Systems Through Distributed Learning and Energy Optimization

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Introduction

Spiking Neural Networks (SNNs) utilize biologically-inspired spikes for energy-efficient communication and computation.

Distributed Learning trains multiple models and then merges them.

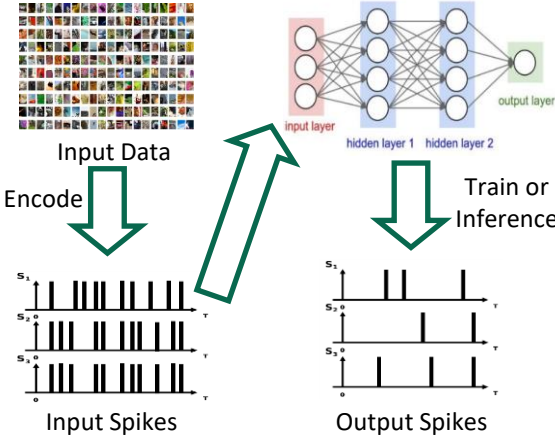
Carbon Neutral Computing Covers SNN energy consumption by harvested energy, such as solar energy.

Challenges:

- The amount of energy harvested fluctuates by weather etc.
- Enforcing energy consumption within energy harvested by reducing model parameters leads to performance loss.

Objective: Achieving carbon neutrality in each node of distributed learning while minimizing performance loss.

Spiking Neural Networks



Devices

Sensor (Camera etc.)

Collects local data at each node.

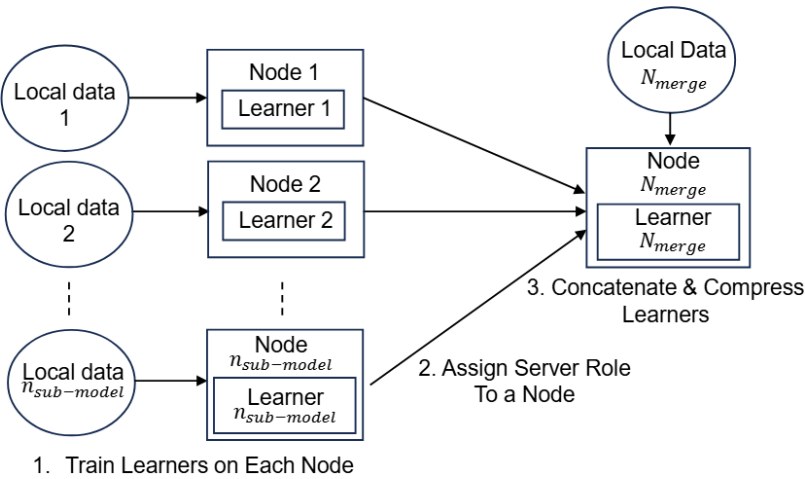
Nodes (IoT devices)

Devices that SNN models are trained. Low computational ability but can be embedded in the local environments.

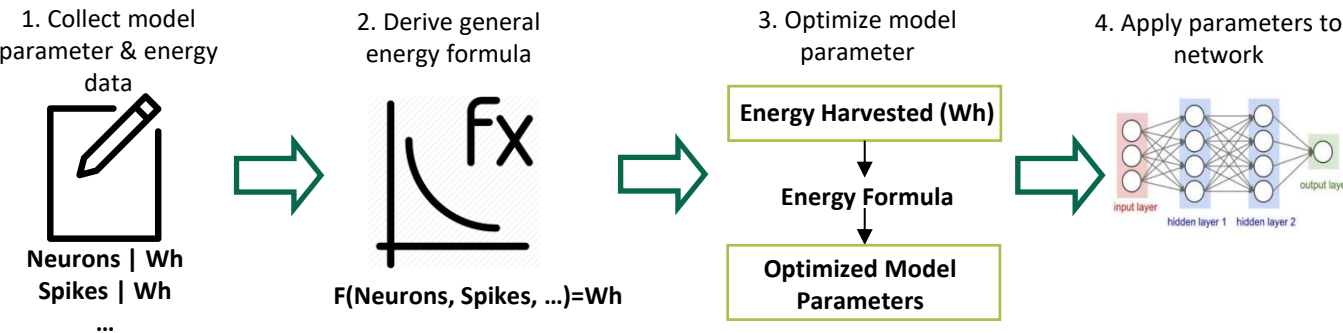
Server (Node of server role)

Collects the pre-processed weights from the nodes, compress them and create the final model.

Distributed Learning Architecture



Energy Optimization



Preliminary Results

Optimizing 70 (neurons) x 10 (SNN models [1]) Training

Weather	Energy Reduction	Accuracy
Sunny	0.00%	85.36%
Cloudy	7.60%	84.26%
Rainy	88.21%	72.79%

References

[1] P. U. Diehl and M. Cook, "Unsupervised learning of digit recognition using spike-timing-dependent plasticity," Frontiers in computational neuroscience, vol. 9, p. 99, 2015.

[2] Hanyu Yuga and Khanh N. Dang, "EnsembleSTDP: Distributed in-situ Spike Timing Dependent Plasticity Learning in Spiking Neural Networks", 17th IEEE MCSoc 2024 (accepted for publication).